Math Kangaroo Practice

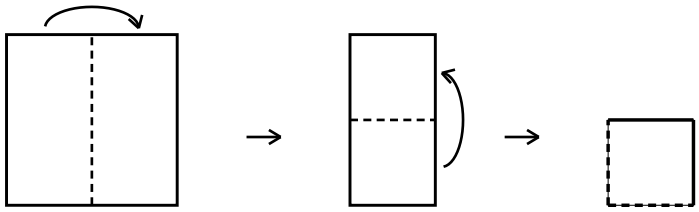
Grade 3-4 · 20 questions · harder · seed 36527

Part A: Each question is worth 3 points

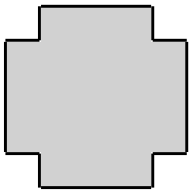
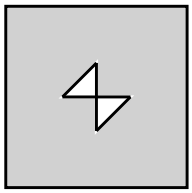
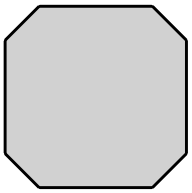
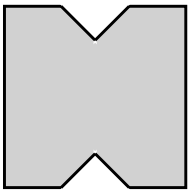
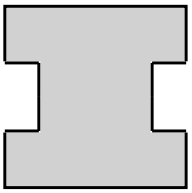
1. Fred has the cards 2, 3, 5 and 7. Fred must use every card exactly once, with +, –, ×, ÷ and brackets.
Which of these numbers can Fred get?

A. 50 ***B.*** 55 ***C.*** 56 ***D.*** 59 ***E.*** 61

2. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square is cut off, and the paper is unfolded.



Which design **cannot** be made this way?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

3. Which number has the same sum of digits as 4641?

A. 4442 ***B.*** 4514 ***C.*** 4731 ***D.*** 4732 ***E.*** 4930

Part B: Each question is worth 4 points

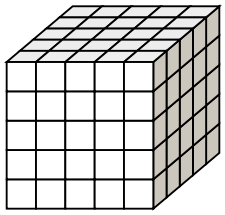
4. Ana has a rope 2 m long and cuts it into pieces that are each 2 dm long.
How many pieces does Ana get?

A. 100 ***B.*** 13 ***C.*** 12 ***D.*** 11 ***E.*** 10

5. The odometer of a car shows 278267 km. Some of its digits are the same.
How many kilometres ago did the odometer last show a number with all different digits?

A. 72 ***B.*** 71 ***C.*** 70 ***D.*** 61 ***E.*** 1

6. A brick 5 cubes long, 5 cubes deep and 5 cubes tall is made of small cubes. It is painted on the outside, also underneath, and then taken apart.



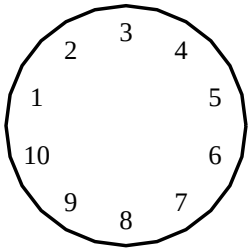
How many small cubes have exactly 2 painted faces?

A. 54 ***B.*** 44 ***C.*** 40 ***D.*** 38 ***E.*** 36

7. Tom was born on October 28, 2004 and Ben on November 21, 2004.
By how many days is Tom older than Ben?

A. 25 ***B.*** 24 ***C.*** 23 ***D.*** 18 ***E.*** 7

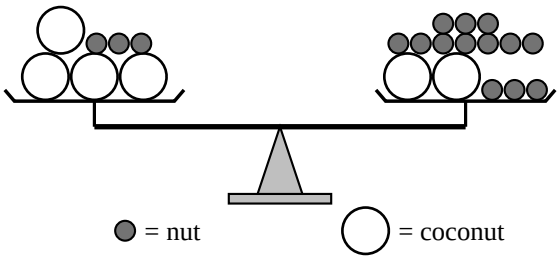
8. A round cake with the numbers 1 to 10 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



What is the largest of these sums?

A. 29 ***B.*** 28 ***C.*** 27 ***D.*** 26 ***E.*** 18

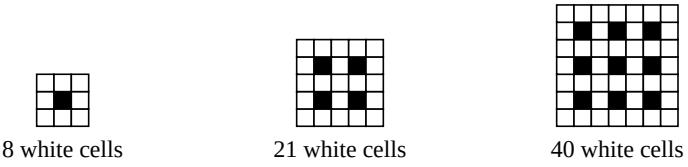
9. The balance is level: 4 coconuts and 3 nuts on one plate, 2 coconuts and 13 nuts on the other.



One coconut weighs the same as how many nuts?

A. 5 nuts ***B.*** 6 nuts ***C.*** 10 nuts ***D.*** 13 nuts ***E.*** 16 nuts

10. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many white cells will the 7th diagram have?



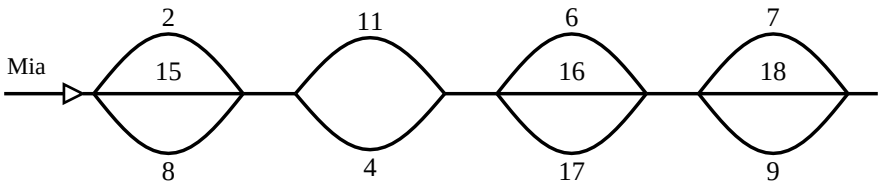
8 white cells 21 white cells 40 white cells

A. 225 ***B.*** 176 ***C.*** 147 ***D.*** 141 ***E.*** 133

11. Which word has exactly 2 different letters in common with TRIANGLE?

A. ELEPHANT ***B.*** PIRATE ***C.*** DINOSAUR ***D.*** CHOCOLATE ***E.*** MONKEY

12. Mia walks along the road from left to right and adds up the numbers along the way.



What is the biggest total Mia can get?

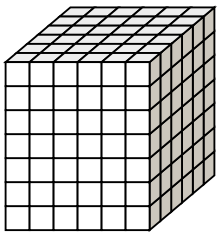
A. 113 ***B.*** 67 ***C.*** 66 ***D.*** 61 ***E.*** 19

13. Sara threads 17 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 6 more blue beads than big red beads. How many small red beads are there?

A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 2

Part C: Each question is worth 5 points

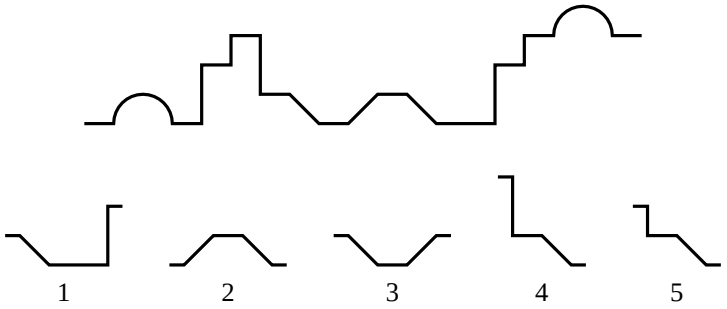
14. Leo builds a brick 6 cubes long, 6 cubes deep and 7 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



How many blue cubes are there?

A. 4 ***B.*** 11 ***C.*** 12 ***D.*** 36 ***E.*** 240

15. Far away we see the skyline of a castle.



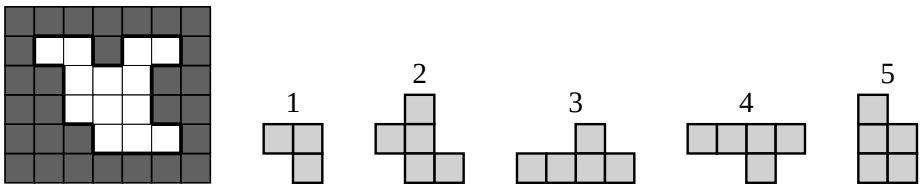
How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

16. Anna paints a flag with three stripes: top, middle and bottom. Anna has 6 colours (white, orange, blue, green, yellow and black). All three stripes must be different colours.
How many different flags can Anna paint?

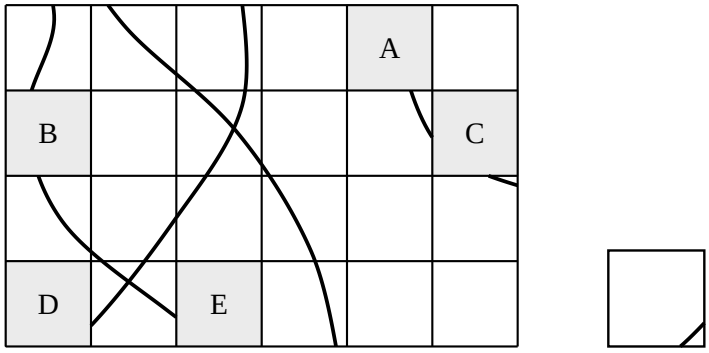
A. 216 ***B.*** 120 ***C.*** 30 ***D.*** 18 ***E.*** 15

17. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+4 ***C.*** 1+2+5 ***D.*** 1+3+5 ***E.*** 2+3+5

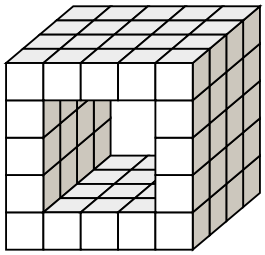
18. Some paths go across a park. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

19. Sara's tunnel is 4 cubes long. The hole goes straight through. Sara makes it longer by adding more cubes at the back in the same pattern, until it is 6 cubes long.



How many cubes does Sara have to add?

A. 10 ***B.*** 16 ***C.*** 28 ***D.*** 32 ***E.*** 96

20. Leo types the numbers from 1 to 100 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,100.

How many keys does Leo press (each digit and each comma is one key)?

A. 300 ***B.*** 298 ***C.*** 292 ***D.*** 291 ***E.*** 290

Answer Key

1.	**B**	**2.**	**B**	**3.**	**C**	**4.**	**E**	**5.**	**B**	**6.**	**E**
7.	**B**	**8.**	**B**	**9.**	**A**	**10.**	**B**	**11.**	**E**	**12.**	**D**
13.	**A**	**14.**	**C**	**15.**	**D**	**16.**	**B**	**17.**	**C**	**18.**	**D**
19.	**D**	**20.**	**D**								

Solutions

1. B (55) — (2 × 7 – 3) × 5 = 55. None of the other numbers can be made. [2002 Q21 · which_can]

2. B — The cut goes through four layers, so it appears four times, mirrored over both folds. Design B is impossible: it is not the same on all four sides, so it is not mirrored over both folds. [2007 Q24 · which_cannot]

3. C (4731) — 4 + 6 + 4 + 1 = 15 and 4 + 7 + 3 + 1 = 15. The others add up to 14 or 16. [2007 Q4 · which_same_sum]

4. E (10) — 2 m = 200 cm and 2 dm = 20 cm. 200 ÷ 20 = 10. [2003 Q10 · pieces]

5. B (71) — Counting down from 278267, the first number with all different digits is 278196: 278267 – 278196 = 71. [2002 Q24 · last_time]

6. E (36) — On each edge, the cubes between the two corners: the brick has 4 edges of each length 5, 5, 5, so 4 × (3 + 3 + 3) = 36. [2007 Q17 · painted_brick]

7. B (24) — October has 31 days, so from October 28 to the end of the month is 3 days, then 21 more days to November 21, 2004: 24 days. [2007 Q12 · days_between]

8. B (28) — All the numbers add up to 55. Two consecutive numbers that add up to 55: 27 + 28 = 55. The largest is 28. [2002 Q11 · largest_sum]

9. A (5 nuts) — Take 2 coconuts and 3 nuts off both plates. Left: 2 coconuts = 10 nuts, so one coconut = 10 ÷ 2 = 5. [2002 Q9 · both_sides]

10. B (176) — Diagram number k is (2k + 1) × (2k + 1) with k × k black cells. The 7th: 15 × 15 – 7 × 7 = 225 – 49 = 176. [2007 Q21 · far_diagram]

11. E (MONKEY) — MONKEY shares E and N with TRIANGLE. PIRATE shares 5; CHOCOLATE shares 4; DINOSAUR shares 4; ELEPHANT shares 5. [2007 Q3 · which_word]

12. D (61) — Take the bigger number at every fork: 15 + 11 + 17 + 18 = 61. [2007 Q1 · largest_sum]

13. A (7) — There are 9 red and 8 blue beads. Big red: 8 – 6 = 2. Small red: 9 – 2 = 7. [2003 Q17 · necklace]

14. C (12) — Each length loses two red cubes at each end, so it drops by 4: 2 × 2 × 3 = 12. [2003 Q9 · thick_wall]

15. D (4) — Pieces 1, 2, 3, 4 fit. Not: 5 (its step is shorter than in the line). [2002 Q7 · how_many_fit]

16. B (120) — Top stripe: 6 choices, middle: 5, bottom: 4. 6 × 5 × 4 = 120. [2002 Q22 · stripes]

17. C (1+2+5) — The empty area has 13 squares, so the three sizes must add up to 13. Only 1 + 2 + 5 also fit together in the right shape. [2003 Q18 · three_pieces]

18. D (gap D) — The square has top: no line, right: 1 line, bottom: 1 line, left: no line. Only gap D has lines reaching it at exactly those places. [2002 Q1 · which_gap]

19. D (32) — Each slice of the tunnel has 5 × 5 – 3 × 3 = 16 cubes. 2 new slices: 16 × 2 = 32. [2002 Q20 · longer_tunnel]

20. D (291) — Digits: 9 × 1 + 90 × 2 + 1 × 3 = 192. Commas: one fewer than the numbers, 99. 192 + 99 = 291. [2007 Q23 · with_commas]